

# What happened when RustRed tried K6

Keeping the hour-long legacy control separate from the new GPLU/SplReD test

## K6 is not closed yet

The first new modular sweep inspected all six root-sector representatives at depth two in about 2.14 seconds. That was not a closing artifact. A later targeted diagnostic did admit and commit one exact generic-root owner for 011111 from its 136-row trace. It also produced one unresolved guard child, while four other roots still had no depth-two modular hit. An audit additionally found a carrier mismatch in the owner-cover delta, so its geometric coverage observation awaits a corrected rerun.

### Legacy control

1:00:18 wall time  
about 119.5 CPU minutes  
about 752 MiB resident memory  
no completed report or artifact

### New root diagnostic

2.137 s total  
6 symmetry-orbit roots  
2 modular hits, 4 bounded misses  
0 authenticated rules, 0 closed roots

The two measurements exercise different engines and must never be merged into one speedup claim.

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## 1. Executive answer

The present evidence answers three different questions:

| Question                                                                  | Answer | Evidence                                                                                                                                                                                         |
|---------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Can the new front end scan a useful finite K6 source window quickly?      | Yes    | All six root representatives, four probes each, and all signed-L1 translations through depth two completed in 2.137 seconds with an early exact-support cap.                                     |
| Did it discover any target-producing modular dependencies?                | Yes    | It found candidates for sectors 001011 and 011111, at the second configured probe.                                                                                                               |
| Did RustRed produce even one new authenticated K6 recurrence in this run? | No     | The dependency traces had 560 and 136 translated source rows. The cap of eight stopped them before exact symbolic materialization.                                                               |
| Has a later targeted diagnostic admitted an exact K6 recurrence?          | Yes    | The 011111 generic-root trace was admitted exactly and one owner was committed. Its guard child remains pending, and the audited carrier mismatch makes the reported cover delta provisional.    |
| Is the complete K6 family closed?                                         | No     | The original sweep left every sector <code>OwnerFree</code> . The later single-owner result neither discharges its exceptional guard nor the four bounded misses, and no global artifact exists. |

### The central performance finding

For this bounded experiment, modular discovery is not the slow part. The first candidate in 001011 is algebraically far too large to be a useful “compact” exact lift: its dependency closure contains 560 rows. An uncapped attempt on that same first orbit ran for more than seven minutes and exceeded 700 MiB before it was interrupted. The comparison strongly implicates exact compact-frame construction/materialization and rational-polynomial expression swell. It does not yet prove a finer phase attribution because that interrupted run had no internal stage timestamps.

There is a second, independent issue. For four orbit representatives, the depth-two search produced no modular target dependency at any of four probes. Those are bounded misses, not mathematical no-go results. They say that this ordering and this small translation window did not reveal a relation. They motivate fair deeper shells, alternative orderings, source preconditioning, and independent probe scheduling.

## 2. What K6 means here

RustRed writes the three-loop complete single-scale vacuum family as

$$I(n_1, n_2, n_3, n_4, n_5, n_6).$$

The six integer coordinates describe propagator or irreducible-numerator powers in a complete scalar-product basis. Hence the shorthand K6: this is a six-coordinate family, not a six-loop calculation. The present pressure case is a three-loop family.

A sector mask such as 001011 records which of the six lines are active. With one-based positions, 001011 means that coordinates 3, 5, and 6 are active. Its corner integral is

$$I(0, 0, 1, 0, 1, 1).$$

The authenticated graph symmetry groups the non-scaleless full-rank sectors into six orbits:

| No. | Representative | Orbit size | Active lines | Meaning in this experiment                   |
|-----|----------------|------------|--------------|----------------------------------------------|
| 1   | 001011         | 12         | 3            | root searched; large modular candidate found |
| 2   | 001101         | 4          | 3            | root searched; no depth-two hit              |
| 3   | 001111         | 12         | 4            | root searched; no depth-two hit              |
| 4   | 011110         | 3          | 4            | root searched; no depth-two hit              |
| 5   | 011111         | 6          | 5            | root searched; modular candidate found       |
| 6   | 111111         | 1          | 6            | root searched; no depth-two hit              |

“Full rank” here excludes sectors already removed by the elementary scaleless- loop proof. It is not an assertion that every listed integral is analytically nonzero. Nor are these six masks six separate hard-coded topologies: they are canonical representatives under the authenticated slot action.

## 2.1. Sector coordinates and the unresolved rays

It is useful to replace signed powers by nonnegative sector coordinates  $x_i$ :

$$x_i = \begin{cases} n_i - 1 & \text{if line } i \text{ is active} \\ -n_i & \text{if line } i \text{ is inactive.} \end{cases}$$

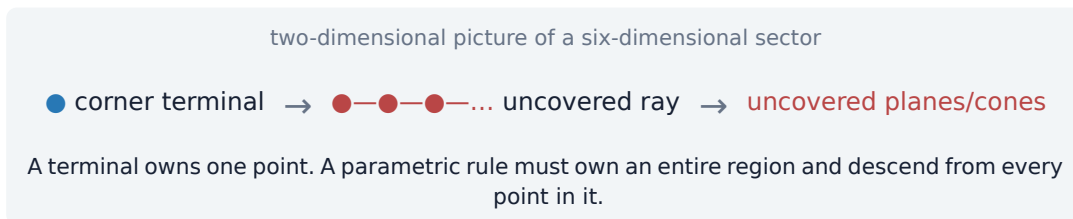
The sector corner is  $x = 0$ . A rule that owns a positive-dimensional region should reduce every lattice point in that region to strictly lower points. A single explicit terminal at the corner does not do this.

For 001011, two elementary rays are

$$R_{\text{active}} = \{I(0, 0, 1 + t, 0, 1, 1) : t \in \mathbb{N}\},$$

$$R_{\text{numerator}} = \{I(-t, 0, 1, 0, 1, 1) : t \in \mathbb{N}\}.$$

The first raises an active power; the second deepens an inactive numerator. There are analogous coordinate rays in all six directions, plus mixed planes and cones. Before an exact rule is admitted, all of these remain part of the unowned positive-dimensional region.



The diagnostic report calls each remaining root `OwnerFree` and reports one uncovered box with `uncovered_finite=false`. In this vocabulary, “finite” means that the residual has collapsed to finitely many fixed terminal keys, not merely that machine-representable i64 coordinates impose a remote safety endpoint. Mathematically, the family still contains unbounded ray-like directions. Because no candidate crossed the exact authority boundary, the run closed none of them.

The complement is a cone-like lattice region, so there is not a short finite list of “the rays.” The following table exhibits one coordinate ray inside each still-uncovered orbit root. Every  $t \in \mathbb{N}$  is a different integral, and mixed changes of two or more powers give further planes and cones.

| Orbit  | One explicit unresolved ray | Root result      |
|--------|-----------------------------|------------------|
| 001011 | $I(0, 0, 1 + t, 0, 1, 1)$   | modular hit only |
| 001101 | $I(0, 0, 1 + t, 1, 0, 1)$   | bounded miss     |
| 001111 | $I(0, 0, 1 + t, 1, 1, 1)$   | bounded miss     |
| 011110 | $I(0, 1 + t, 1, 1, 1, 0)$   | bounded miss     |
| 011111 | $I(0, 1 + t, 1, 1, 1, 1)$   | modular hit only |
| 111111 | $I(1 + t, 1, 1, 1, 1, 1)$   | bounded miss     |

For masks with inactive positions there are also numerator rays, obtained by replacing one inactive zero by  $-t$ . Even the two “modular hit” rows in the table remain uncovered: proposal evidence does not install an owner.

### 3. Two experiments, not one

The most important interpretive rule for these measurements is that the hour-long run and the 2.137-second run are different programs of work.

#### A · Legacy owner-cover control

campaign run-waves  
old ProbeCampaignAdapter path  
bounded atomic orbit waves  
no SplReD target stream  
no useful completion report in one hour

≠

#### B · New SplReD root diagnostic

one equality-root attempt per orbit  
signed-L1 source shells  
dual modular reducers  
direct dependency traces  
exact lift stopped at eight rows

Therefore:

- the one-hour control does not measure GPU/SplReD throughput;
- the 2.137-second sweep does not measure full K6 closure;
- dividing one time by the other would be a meaningless speedup number;
- neither run produced a cold-loadable K6 artifact.

#### 3.1. Experiment A: the hour-long legacy control

The command requested four workers and launched the previous bounded full-rank-wave campaign. The first canonical wave has width two, so only two tasks were available. The process consequently stayed near 198% aggregate CPU instead of saturating four cores.

| Observed quantity             | Value           | What it establishes                                                                                             |
|-------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Wall time before interruption | 1:00:18         | The first useful completion boundary was not reached in one hour.                                               |
| CPU time                      | about 119.5 min | Two cores were effectively occupied for most of the run.                                                        |
| Aggregate CPU                 | about 198%      | Consistent with the first wave’s width of two, despite <code>n_cores=4</code> .                                 |
| Resident memory               | about 752 MiB   | The old campaign retained a sizeable working set, but did not show runaway growth during the later observation. |
| Completed stdout report       | none            | No per-phase or per-orbit result crossed the reporting boundary.                                                |
| Written artifact              | none            | No closure result exists from this run.                                                                         |

Operational snapshots showed roughly 667 MiB after about 74 seconds and a working set near 752 MiB by approximately 8.5 minutes. At that point the process had accumulated about 17 CPU minutes, again consistent with two busy workers. It continued without a task-completion report until manually stopped.

#### What happened during that hour

Two first-wave legacy tasks ran continuously inside the old owner-cover campaign. They did not finish far enough to emit the expected report or artifact. The external process observations establish total time, CPU use, concurrency, and memory. They do not reveal how much of the hour belonged to modular sampling, exact algebra, cover compilation, or another internal phase. Any finer attribution would be invented.

This control was useful precisely because it exposed two engineering problems: coarse wave width underuses the requested worker budget, and progress emitted only at a large task boundary makes a long run opaque. It is not evidence against the mathematics of case-directed completion.

### 3.2. Experiment B: the new six-orbit root diagnostic

The new release diagnostic was designed to answer a narrower question quickly: what happens at the first generic equality case of every K6 orbit when the SplReD target stream is actually exercised?

Its fixed inputs were:

- nine freshly generated ordinary K6 IBP source rows;
- canonical target  $I(n)$ , i.e. zero structural shift;
- signed-L1 translations through depth two;
- four case-valid finite-field probes;
- a fresh exact predecessor/owner ledger for each orbit;
- at most 32 post-hit streamed rows and eight distinct compact-lift attempts;
- an ingress cap of eight selected exact rows;
- a release build, with compilation excluded from the reported 2.137 seconds.

For six translation coordinates, the number of offsets is

$$1 + 12 + \left(12 + 4 \binom{6}{2}\right) = 1 + 12 + 72 = 85.$$

There are nine ordinary sources, so one complete depth-two probe streams

$$85 \times 9 = 765 \text{ rows.}$$

Four complete no-hit probes therefore stream  $4 \times 765 = 3060$  rows. This arithmetic is visible in the measured census and is a useful check that the fair shells really completed.

#### 3.2.1. The four probes

| No. | Prime      | $d$ | Index chart point    |
|-----|------------|-----|----------------------|
| 1   | 998244353  | 29  | (0, 0, 0, 0, 0, 0)   |
| 2   | 998244353  | 43  | (1, 1, 1, 1, 1, 1)   |
| 3   | 1000000007 | 29  | (2, 3, 5, 7, 11, 13) |
| 4   | 1000000009 | 43  | (13, 11, 7, 5, 3, 2) |

Using the same prime in the first two probes is intentional here because the index and dimension specialization differs. A production fairness schedule must eventually vary both points and primes rather than growing source depth forever at one unlucky specialization.

## 4. How the modular GPLU-style search is applied

Each translated ordinary identity has the form

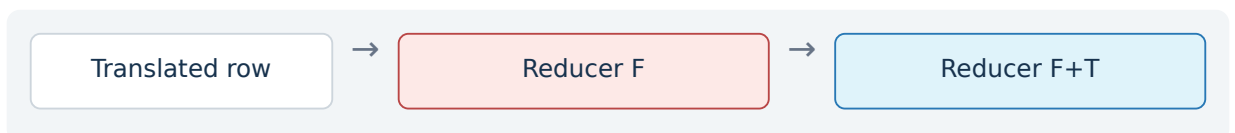
$$\sum_s c_{s(d,n+a)} I(n + a + s) = 0,$$

where  $a$  is the source translation and  $s$  is a structural shift inside one ordinary IBP row. At a modular probe, RustRed evaluates  $c_{s(d,n+a)}$  directly in a prime field. It does not first construct the exact shifted rational polynomial.

The exact owner snapshot and ordering classify every integral column into three roles:

|                                                                                              |                                                                                                                               |                                                                                                                         |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Target</b><br><br>The stable logical column for $I(n)$ that this case is trying to solve. | <b>Allowed</b><br><br>A term already justified as lower or previously owned. It need not be cancelled by the new combination. | <b>Forbidden</b><br><br>A non-target term that is not yet legal on the right-hand side. The combination must cancel it. |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|

Allowed columns are omitted from discovery. Every row is streamed once into two incremental Symbolica sparse reducers:



$c_F$  on forbidden  
columns  
plus target coeffi-  
cient  $c_T$

forbidden columns only  
measures  $\text{rank}(F)$

same forbidden columns  
plus stable target  
measures  $\text{rank}(F|T)$

A target candidate exists when adding the target column changes the relevant rank/pivot condition:

$$\text{rank}(F \mid T) > \text{rank}(F).$$

Equivalently, some linear combination of the streamed identities cancels all forbidden columns while retaining a nonzero target coefficient. The reducer's lower-factor pattern records direct predecessor edges. RustRed walks those edges backward only after a hit, retaining a dependency DAG rather than expanding every transitive row combination throughout the stream.

#### Terminology precision

This document uses “GPLU” as the project shorthand for the planned Gilbert–Peierls/LU-pattern dependency workflow. The pinned public Symbolica reducer provides incremental sparse reduction and an L pattern, but its forward path still uses a dense scratch row. It is therefore not yet a textbook reachability-driven Gilbert–Peierls kernel. That distinction is important: dense forward scans and dense dependency patterns can contribute to large traces even when the input identities are sparse.

### 4.1. Why a modular hit is not an IBP rule

The modular computation answers only an existence question at one prime and one point. A valid parametric recurrence still requires:

1. regenerating precisely the selected translated ordinary sources;
2. repeating the candidate construction in exact Symbolica rational-polynomial arithmetic;
3. proving the resulting identity has zero exact residual;
4. extracting every denominator and leading-coefficient guard;
5. proving all remaining integrals are strictly lower under the persisted ordering;
6. splitting and rescheduling every exact exceptional equality case;
7. compiling owners and proving that the final uncovered complement is finite.

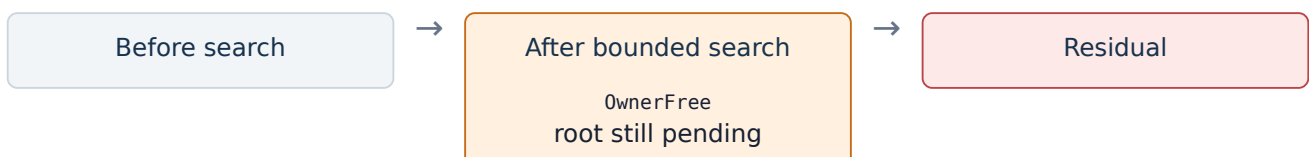
Only then may the rule enter an immutable artifact. This strict boundary is why the measured candidates close no rays yet.

## 5. The measured SpIReD results

| Orbit  | Size | Result | Probe / rows | Trace rows | Elapsed  | Published |
|--------|------|--------|--------------|------------|----------|-----------|
| 001011 | 12   | hit    | 2 / 1456     | 560        | 106.6 ms | no        |
| 001101 | 4    | no hit | 4 / 3060     | —          | 296.8 ms | no        |
| 001111 | 12   | no hit | 4 / 3060     | —          | 482.6 ms | no        |
| 011110 | 3    | no hit | 4 / 3060     | —          | 549.5 ms | no        |
| 011111 | 6    | hit    | 2 / 1000     | 136        | 27.8 ms  | no        |
| 111111 | 1    | no hit | 4 / 3060     | —          | 642.0 ms | no        |

The six per-orbit times sum to about 2.105 seconds. Roughly 32 milliseconds of remaining setup/reporting overhead gives the measured all-orbit total of 2.137 seconds.

Every final ledger had the same authority status:



one declared corner terminal  
no rule owner

one positive-dimensional  
uncovered box  
not terminal-finite

### 5.1. What relation did it “stumble on”?

For 001011, the stream found a finite-field dependency after an aggregate 1456 rows, during the second configured probe. Walking the direct predecessor edges showed that the candidate depends on 560 translated ordinary source requests. For 011111, the corresponding numbers were 1000 streamed rows and 136 dependency rows.

These are precise candidate identities, but they are not exact parametric IBPs. RustRed does not yet have authenticated coefficients, a denominator, a right-hand side, guards, or a descent proof for either one. Consequently there is no honest formula to print in this note. Writing a plausible-looking recurrence from the modular residue would fabricate authority that the run did not produce.

#### The exact-row cap did its job

The diagnostic cap was eight selected rows. Since  $560 > 8$  and  $136 > 8$ , both candidates were stopped before building the exact selected-source epoch. The cap is not a claim that no valid relation exists. It converts a likely symbolic blow-up into a fast, typed, reproducible incomplete result.

#### Behavior after this measured sweep

The measured table above is retained as historical evidence: that version of the root diagnostic returned when its oversized first candidate hit the cap. The now-tested target runner instead treats an oversized support as an ordinary candidate-local inconclusive result and continues its bounded post-hit window and alternative-support search. It still constructs no exact epoch for the oversized trace and grants no authority. New ordering studies must therefore be compared with each other, not silently substituted into the historical 2.137-second census.

### 5.2. Follow-up: one exact 011111 root owner

A subsequent targeted release diagnostic deliberately raised the exact limit for the smaller 011111 witness. This time the 136-row dependency trace passed through exact materialization and admission. The literal census was:

| Quantity                        | Observed                     | Meaning                                                                                                           |
|---------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Internal target-run time        | 145.202 ms                   | Discovery, exact admission, guard extraction, and owner commit inside the measured target run.                    |
| Setup-inclusive time            | about 171 ms                 | The surrounding one-orbit diagnostic, excluding compilation.                                                      |
| Source requests / streamed rows | 1394 / 1064                  | The bounded source stream stopped after the admitted candidate rather than exhausting four full depth-two probes. |
| Modular hits / lift requests    | 1 / 9                        | One target-producing modular event; nine compact-lift proposals were considered in its bounded window.            |
| Exact admissions                | 1                            | The 136-row trace produced a valid exact algebraic recurrence; this is no longer merely modular evidence.         |
| Guards / committed owners       | 1 / 1                        | The generic root gained one owner and one exceptional guard case was proposed.                                    |
| Ledger snapshot                 | rev=1, owners 1, terminals 1 | The reported status was <code>Incomplete(GuardIncomplete)</code> , not closed.                                    |



| Quantity        | Observed                                    | Meaning                                   |
|-----------------|---------------------------------------------|-------------------------------------------|
| Residual report | 1 uncovered box,<br>1 guard-incomplete case | One proposed guard child remains pending. |

### A genuine exact step, still not K6 closure

The exact algebraic admission is valid: RustRed rebuilt the selected ordinary sources, obtained an exact recurrence, passed admission, and committed an owner. It does not discharge the coefficient-zero branch of that rule. The ledger therefore correctly reports `GuardIncomplete`; a generic rule cannot own the locus on which its leading coefficient vanishes.

An adversarial audit found a separate geometric bookkeeping defect in this diagnostic: the owner-free baseline used for its delta comparison was taken on the full orthant rather than on the source-safe carrier. This does not undo the exact algebraic recurrence, but it makes the observed owner-cover delta and any claim of geometric shrinkage provisional. The carrier comparison must be fixed and the run repeated before treating that delta as authoritative.

A serial equality-case driver is also being connected to this root path. Its current diagnostic probe portfolio does not yet match the intended case-valid probe schedule; that wiring mismatch is being fixed. Until the corrected driver reruns, its bounded carrier is useful for exercising control flow but cannot prove global closure of an unbounded sector.

### 5.3. Thirty-ordering bounded screen

After oversized supports became ordinary candidate-local inconclusive results, the same release executable screened 30 deterministic pinned global ordering representatives. Every run used depth two, four probes per root, exact-row cap eight, post-hit window 64, all six root orbits, and a single test thread. A priority list below is the persisted rank by coordinate slot, not a K6 topology label.

Every ordering gave the same coverage result:

- 001011 had one modular hit;
- 011111 had six modular hits across the bounded continuation;
- 001101, 001111, 011110, and 111111 had no modular hit after 3060 rows each;
- no cap-eight run admitted a rule; every ledger remained `OwnerFree` with one positive-dimensional uncovered box.

The detailed candidate census falls into three compact profiles:

| Code | 001011                              | 011111                              |
|------|-------------------------------------|-------------------------------------|
| A    | rows=3050, hits=1, lift_requests=9  | rows=2228, hits=6, lift_requests=22 |
| B    | rows=2992, hits=1, lift_requests=15 | rows=2228, hits=6, lift_requests=22 |
| C    | rows=3050, hits=1, lift_requests=9  | rows=2270, hits=6, lift_requests=14 |

| Rank by slot | Total (ms) | Profile |
|--------------|------------|---------|
| 0,1,2,3,4,5  | 3069.266   | A       |
| 0,1,2,3,5,4  | 2774.234   | B       |
| 0,1,2,4,3,5  | 2928.470   | A       |
| 0,1,2,4,5,3  | 2781.875   | B       |
| 0,1,2,5,3,4  | 2745.850   | A       |
| 0,1,2,5,4,3  | 2728.770   | B       |
| 0,1,3,2,4,5  | 2786.279   | A       |
| 0,1,3,2,5,4  | 2804.341   | B       |
| 0,1,3,4,2,5  | 2588.342   | A       |
| 0,1,3,4,5,2  | 2539.085   | A       |
| 0,1,3,5,2,4  | 2540.823   | A       |
| 0,1,3,5,4,2  | 2532.825   | A       |

| Rank by slot | Total (ms) | Profile |
|--------------|------------|---------|
| 0,1,4,2,3,5  | 2653.519   | A       |
| 0,1,4,2,5,3  | 2605.839   | A       |
| 0,1,4,3,2,5  | 2678.323   | A       |
| 0,1,4,3,5,2  | 2814.695   | A       |
| 0,1,4,5,2,3  | 2803.082   | B       |
| 0,1,4,5,3,2  | 2679.163   | A       |
| 0,1,5,2,3,4  | 2716.180   | B       |
| 0,1,5,2,4,3  | 2625.718   | A       |
| 0,1,5,3,2,4  | 2615.171   | B       |
| 0,1,5,3,4,2  | 2775.539   | A       |
| 0,1,5,4,2,3  | 2667.850   | B       |
| 0,1,5,4,3,2  | 2788.268   | A       |
| 0,2,3,4,5,1  | 2656.890   | C       |
| 0,2,3,5,4,1  | 2749.653   | C       |
| 0,2,4,3,5,1  | 2713.258   | C       |
| 0,2,4,5,3,1  | 2649.527   | C       |
| 0,2,5,3,4,1  | 2615.716   | C       |
| 0,2,5,4,3,1  | 2765.603   | C       |

Across these single observations, the minimum was 2532.825 ms, the median 2714.719 ms, the mean 2713.138 ms, and the maximum 3069.266 ms. Peak resident memory ranged from 9236 to 15376 KiB, with a median of 12312 KiB. The fastest single observation was rank-by-slot 0,1,3,5,4,2, but these sequential runs were neither repeated nor isolated from shared-host load. It is therefore a screening result, not evidence of a statistically meaningful “winning” ordering.

The built-in natural ordering, which has no explicit coordinate-priority overlay, is distinct from the explicit identity list 0,1,2,3,4,5. Its separate cap-eight control took 2541.666 ms. The previously suggested priority 5,3,4,2,0,1 took 2926.693 ms, and the explicit reverse priority 5,4,3,2,1,0 took 2538.435 ms. They too changed neither bounded hit coverage nor rule admission.

Three cap-16 checks gave the following internal totals:

| Ordering         | Total       | Outcome          |
|------------------|-------------|------------------|
| built-in natural | 2369.801 ms | 0 rules admitted |
| 0,1,3,5,4,2      | 2505.045 ms | 0 rules admitted |
| 5,3,4,2,0,1      | 2534.059 ms | 0 rules admitted |

Raising the cap from eight to sixteen therefore exposed no small admissible witness in these checks. The continuation diagnostic does not yet print the support size of every rejected candidate, so the matrix cannot honestly name its smallest trace. The literal known first-hit traces remain 560 and 136 from the earlier measurement; the later exact 011111 run is the evidence that the 136-row trace can in fact be materialized and admitted.

#### What the ordering screen says

Ordering already changes row rejection, fill chronology, and the number of compact-lift proposals. At this shallow shell it does not rescue any of the four modular misses, nor find a cap-small owner for the two hit roots. The 30 representatives form a principled deterministic portfolio, but keeping the six canonical orbit representatives fixed means this is not an exhaustive proof over all relative orderings. Deeper fair shells, preconditioning, and trace-quality scoring remain necessary.

The useful discovery is therefore structural:

- 001011 does have a target-producing dependency in the tested modular window, but the first dependency path is enormous;

- 011111 also has one, and its first path is about four times smaller; the follow-up exact diagnostic has now admitted that 136-row generic recurrence, but its guard child remains open;
- the present row ordering and pivot chronology are selecting poor exact witnesses, especially in 001011;
- searching only until the first hit is not an adequate performance policy.

#### 5.4. The four no-hit roots

For 001101, 001111, 011110, and 111111, every configured probe consumed all 3060 rows without a target rank change. No exact stage was attempted.

This does not imply that these sectors lack a parametric recurrence. A finite exact minor can vanish at an unlucky point or prime, and a valid relation may need a translation outside the depth-two diamond. The correct status is:

##### Bounded discovery miss

“No candidate was found for this ordering within signed-L1 depth two and the four listed specializations.” It is not “this orbit is a master,” “this ray is closed,” or “no relation exists.”

The elapsed times differ because the number and order of dynamically observed forbidden columns, sparse fill, and dense scratch-row work differ by sector. The current report does not expose enough per-row reducer telemetry to attribute the 296–642 ms spread more finely.

## 6. The uncapped first-orbit attempt

Before adding the ingress cap, the same new root diagnostic began with 001011. It did not print the first-orbit result after more than seven minutes, used approximately one CPU core, and grew beyond 700 MiB resident memory. It was interrupted before any rule, owner, report line, or artifact was produced.

The capped rerun later demonstrated that:

1. the modular candidate for this orbit is found in about 0.1 seconds;
2. its exact dependency trace has 560 rows;
3. returning before exact epoch construction avoids the long stall and large working set.

Together, these observations strongly implicate the exact compact-lift path: rebuilding hundreds of exact shifted rows and reducing their multivariate rational-polynomial coefficients can trigger denominator products, repeated GCDs, and rapid intermediate support growth.

##### What is proved, and what is inferred

Proved by the censuses: modular scouting reaches the candidate quickly; the candidate has 560 dependency rows; an early cap makes the attempt return quickly.

Strongly inferred: the uncapped time and memory were spent in compact exact-frame construction/materialization and its Symbolica algebra.

Not yet measured: the division among exact source translation, sparse matrix construction, rational-polynomial row reduction, GCD/cancellation, replay, and guard extraction. Stage-level timers are required before naming one of those subphases as the sole bottleneck.

This distinction matters because “expression swell” is a mechanism, not yet a complete profile. The next diagnostic must report exact input rows and terms, intermediate nonzeros, polynomial terms and degree, GCD work, peak memory, and time around every non-interruptible Symbolica call.

## 7. Where time is currently spent

The evidence supports the following bottleneck map.

| Stage                                                            | Current evidence                  | Interpretation                                                                                                                         |
|------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Structural source preparation and modular coefficient evaluation | fast in bounded sweep             | Included in the 2.137-second all-orbit total. Not the dominant root cost at depth two.                                                 |
| Dual incremental modular reduction                               | fast enough                       | Four complete 3060-row misses take 0.30–0.64 s each. There is optimization room, but no minute-scale problem here.                     |
| Dependency extraction                                            | cheap but produces large output   | The direct L-pattern walk quickly exposes traces of 560 and 136 rows. Trace quality, not extraction latency, is the problem.           |
| Exact compact-frame construction/materialization                 | strongly implicated               | The uncapped 560-row candidate runs for minutes and consumes hundreds of MiB; the preflight cap makes it return in 0.1 s.              |
| Exact replay, guards, and descent                                | reached for the 136-row trace     | The complete admitted target run took 145.202 ms, but its subphases are not timed separately. The 560-row path remains unmaterialized. |
| Exceptional-case fixed point                                     | one child exposed, not discharged | The exact 011111 owner proposed one guard child; the ledger remains <code>GuardIncomplete</code> .                                     |
| Legacy owner-cover control                                       | opaque and slow                   | One hour before its first report boundary; its phases cannot be recovered from external telemetry.                                     |

The immediate optimization target is thus not “make every finite-field operation a little faster.” It is “make the winning exact dependency much smaller, and avoid exact rational-polynomial elimination until that has been achieved.”

## 8. What Gregor’s case-directed strategy says to do next

Gregor’s notes change how the results should be read. The purpose of the finite-field pass is not to obtain the first dependency at any cost. It is to find a small, useful witness for one exact case, recenter it to the canonical target, lift only that witness, and recursively handle the exact loci where its pivot coefficient vanishes.

The K6 measurements make five consequences concrete.

### 8.1. Improve the witness, not only the arithmetic

The first 001011 hit contains 560 rows. Even perfect exact arithmetic cannot make that an attractive starting point. RustRed should compare candidates in a bounded post-hit window, scoring at least:

- dependency-node and edge counts;
- predicted exact polynomial term work;
- forbidden-column fill and pivot density;
- expected guard degree and branch count;
- source translation depth;
- stability across held-out probes.

The existing post-hit lane is the right seam. It should not accept the first rank gain when a slightly later row yields a dramatically smaller dependency.

### 8.2. Precondition ordinary sources without losing provenance

A sector- and ordering-local sparse exact RREF can remove dependent raw source directions before translation. Denominators may be cleared and rows made primitive, while preserving a transformation back to the nine generated ordinary sources for final replay.

This is an acceleration, not new mathematical authority. A raw-source lane must remain a fair fallback because specialization can make a generally useful preconditioned row degenerate on one equality case. The K6 experiment should compare raw and preconditioned streams by hit depth, trace size, modular fill, and exact-lift cost.

### 8.3. Treat ordering as part of the solver

The dependency trace is a consequence of source chronology, integral-column order, and pivot choices. A poor order can turn a short relation into a dense 560-row ancestor closure. RustRed should race a small deterministic, symmetry-reduced ordering portfolio and retain the cheapest exact candidate.

The comparison must remain fair: preferences may reorder sources within a signed-L1 shell, but no source can be starved forever. If a small static portfolio still produces outliers, a bounded bandit/MCTS policy can use early fill and trace-growth signals to abandon bad orders before exact materialization.

### 8.4. Continue the four misses fairly

For the four no-hit orbits, move diagonally through a schedule of increasing translation depth, independent primes, index points, and orderings. Do not run depth 3, then 4, then 5 forever at the same specialization. If a finite exact minor exists, some prime and point should preserve it; if a finite translated witness exists, some shell should contain it.

This is a conditional semi-decision procedure, not a known finite-depth bound. Resource exhaustion must save resumable progress and remain “incomplete.”

### 8.5. Delay exact coefficient construction

Gregor’s fast route relies on modular discovery and rational-polynomial reconstruction. Symbolica does not yet expose the required optimized sparse multivariate rational-function reconstruction API. RustRed should therefore:

- retain a narrow materializer interface;
- use exact Symbolica arithmetic only for genuinely small pruned traces now;
- reject or defer huge traces with typed resource diagnostics;
- plug in Symbolica’s reconstruction materializer when the public API arrives;
- not implement a competing interpolation/CRT/CAS framework internally.

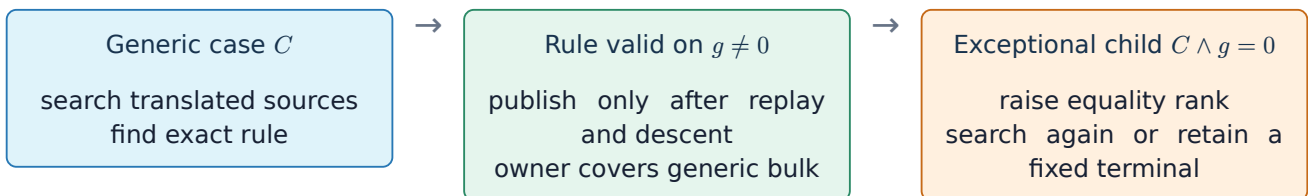
Even after reconstruction exists, a 560-row trace is undesirable. Better ordering and preconditioning remain essential because reconstruction reduces coefficient cost, not combinatorial fill.

## 9. Completing a root is still not completing K6

Suppose a future ordering produces the exact generic rule

$$I(\mathbf{n}) = \sum_j r_{j(d,\mathbf{n})} I(\mathbf{n} + \delta_j).$$

If its leading coefficient contains a factor such as  $n_1$ , the rule applies only where  $n_1 \neq 0$ . The equality face  $n_1 = 0$  must become a new case. On that face, a term that would otherwise enter an inactive-line supersector may vanish because its coefficient is also proportional to  $n_1$ . This coefficient-aware boundary analysis can avoid unnecessary children, but it must be exact.



When guards are coordinate-linear, every nonredundant child fixes another coordinate, so equality rank increases and recursion depth is at most six. Coupled affine guards require the planned exact lattice chart. Non-affine or unsupported exceptional geometry must fail closed.

The production driver must also turn genuinely activating inactive-boundary faces into working cells, while keeping its temporary bounded search envelope separate from the logical equality case. These components are necessary before a root rule can grow into a zero-uncovered artifact.

## 10. A concrete next-run programme

The live measurements and Gregor’s strategy together suggest the following ordered programme.

1

**Add exact-stage telemetry**

Timestamp source regeneration, exact epoch construction, matrix assembly, Symbolica row reduction, replay, guard extraction, descent, and owner compilation. Record rows, terms, nonzeros, polynomial degrees/support, and RSS around native calls. This turns the strong exact-swell inference into a profile.

2

**Shrink the pathological candidate**

Concentrate trace-quality work on the still-pathological 560-row 001011 witness. Compare raw and provenance-preserving preconditioned sources, then score a bounded post-hit ordering portfolio. Retain 011111 as a positive exact-admission regression and seek a smaller trace only if profiling makes its 136-row materialization material.

3

**Deepen the four misses**

Resume signed-L1 shells fairly across new primes and points. Persist prepared structural rows and reducer-compatible progress where possible; restarting depth zero for every probe/depth expansion wastes the chronology.

4

**Consolidate the admitted root rule**

Fix the source-safe-carrier baseline and serial probe wiring, then reproduce the exact 011111 admission through the production driver. Preserve its regenerated-source replay, guard extraction, and strict descent evidence; process the proposed guard child instead of treating the generic owner as root closure.

5

**Run the equality-case fixed point**

Feed guard-zero and inactive-boundary cases into the most-generic-first worklist. Keep positive-dimensional misses pending; only fully fixed leaves may use the explicit nonminimal terminal policy.

6

**Attempt full K6 again**

Run all six orbit roots through the production SplReD driver, cold-write the artifact, reload it in a fresh process, prove zero uncovered cases, and reduce canary integrals. Only this result warrants the phrase "K6 closes."

The full K6 attempt should be repeated after each coherent change to ordering, preconditioning, compact tracing, or exact materialization. A small bounded root sweep remains the fast regression, while the full closure run supplies the real end-to-end pressure.

## 11. What success will look like

A future successful log must contain more than a short runtime and six modular hits. At minimum it must demonstrate:

| Gate            | Required evidence                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------|
| Discovery       | Every positive-dimensional root/child has a finite candidate; modular support is stable under held-out probes. |
| Exact authority | Every rule reconstructs from freshly generated ordinary sources and replays to exact zero.                     |
| Applicability   | All leading coefficients and denominators have explicit guards; every zero locus is discharged.                |

| Gate            | Required evidence                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| Termination     | Every right-hand-side integral is strictly lower under the artifact's persisted order.                                    |
| Coverage        | All six sector orbits and their symmetry images have zero positive-dimensional uncovered complement.                      |
| Terminals       | Only an explicit finite terminal manifest remains; a nonminimal basis is acceptable.                                      |
| Reproducibility | Deterministic artifact identity across supported worker counts; fresh-process load and canary application pass.           |
| Integration     | The shipped artifact drives Vakint's FORM-less RustRed scalar backend through all applicable three-loop acceptance tests. |

### Present status

The new approach has reached one exact algebra milestone, not a closure milestone. It turns the root scan from an opaque hour-scale attempt into a two-to-three-second, orbit-by-orbit diagnosis, and it can admit the 136-row 011111 generic recurrence in about 145 ms internally. Its guard child, four bounded discovery misses, the 560-row 001011 expression-swell path, and the audited carrier/probe wiring defects remain. No K6 artifact can yet be published.

## 12. Reproduction boundary

The diagnostic source lives in `crates/rustred-core/src/foundry/campaign/k6_spired_root_diagnostic.rs` and is marked ignored because it is a release-only K6 pressure test. The intended invocation is:

```
export SYMBOLICA_LICENSE='<valid local licence>'
nix develop --command cargo test --release -p rustred \
  k6_spired_root_attempt_reports_every_full_rank_orbit \
  -- --ignored --nocapture
```

Compilation and dependency download are outside the 2.137-second logical test boundary. The exact-row cap and source depth are diagnostic policy, not production closure settings. Repeating the command after changing either one must record the new configuration alongside time, CPU, peak RSS, row counts, trace sizes, rule admissions, pending cases, and cover status.

The legacy hour-long control used `campaign run-waves` with the autonomous K6 configuration and four requested cores. Its exact command line and temporary output location belong in the run transcript, but its runtime must remain labelled “legacy owner-cover control,” never “GPLU K6.”

## 13. Final plain-language summary

RustRed now has a fast way to ask: “among these translated ordinary IBPs, can a finite-field linear combination isolate my target while cancelling everything that is not already reducible?” For all six K6 root sectors, a small search takes roughly two to three seconds in the measured depth-two configurations. It says “yes” in two sectors and “not yet” in four, across all 30 tested coordinate-priority representatives.

The two “yes” answers need 560 and 136 source rows in the first measured traces. The smaller 011111 trace has now been materialized exactly and committed as a generic-root owner. Its coefficient-zero guard case is still pending, and an audited carrier mismatch makes its geometric delta provisional until rerun. Trying to materialize the larger 001011 trace is still the operation most strongly implicated by the seven-minute, 700-MiB interrupted attempt. A safe cap catches that path before the expensive work begins.

The global status therefore remains incomplete. The audit checkpoint retains the safe distinction between closure of a bounded diagnostic carrier and closure of the full sector, plus the nonparallel serial-probe portfolio. A direct attempt to make the owner-free planner use the finite compiler carrier was reverted

because the existing planner contract deliberately starts from a symbolic unbounded orthant; reconciling those two scopes remains an explicit architectural seam rather than a hidden closure assumption.

Work now pauses at this reproducible checkpoint. The next implementation pass will begin from Gregor's SplReD reference code once it is supplied under `vendor`: use that winning execution path to resolve the carrier contract, discharge the guard child, improve the `001011` witness, and deepen the four misses. Exact inactive-face and equality-case recursion must then reach a fixed point. Only a cold-reloaded artifact with zero positive-dimensional complement establishes K6 closure.